

host_agent

423 buildings with basement flats have been found in Exeter. These buildings have been plotted on a map, showing their locations and key details such as building type, use, number of floors, and area.

- The map artifact is named: exeter_buildings_with_basement_flats_map.
- Each building on the map includes a summary popup and tooltips for more information.

If you would like to view or download the map, or need further analysis or filtering, please let me know!

planning_agent

The area specified is Exeter, so there will be 1 entry for Exeter. The entity to search for is buildings with basement flats. There is no specific condition mentioned regarding proximity or other filters, so we will look for as many entries as possible for buildings with basement flats in Exeter.

output steps: ["Find Exeter 1 area", "Search for buildings with basement flats in Exeter as many search results"]

named_area_agent

The specific area polygon for Exeter has been filtered and saved as 'filtered_exeter_polygon'.

data_analysis_agent

Filtered Exeter polygon: 1 entry found. An artifact named 'filtered_exeter_polygon' has been generated, containing the GeoDataFrame with the polygon(s) specifically for Exeter, filtered by name and relevant description columns.

Filtered buildings with self-contained basement flats: 423 entries. An artifact has been generated with the name 'exeter_buildings_with_basement_flats'. This artifact is a GeoDataFrame of buildings in Exeter where the 'basementpresence_selfcontained' column indicates the presence of a self-contained flat in the basement. It includes all original columns and geometry.

buildings_agent

Search Results: - 423 buildings with self-contained basement flats were found within the area defined by the artifact 'filtered_exeter_polygon'.

Filtered Artifact: - exeter_buildings_with_basement_flats: This artifact contains a GeoDataFrame of buildings in Exeter where the 'basementpresence_selfcontained' column indicates the presence of a self-contained flat in the basement. It includes all original columns and geometry.

If you need further analysis or details about these buildings, let me know!

plotting_agent

Plotted 423 buildings with basement flats in Exeter.

First 5 entries: 1. ID: e8989c92-0a62-4904-a362-6afdf4bbaf7e, Description: Multiple Residential Accommodation, Use: Residential Accommodation, Floors: 2.0, Area (m2): 1136.95 2. ID: ed420f11-e564-430d-9c33-e0f0a3405379, Description: Multiple Residential Accommodation, Use: Residential Accommodation, Floors: 2.0, Area (m2): 42.452 3. ID: d4a4d304-dcdf-42db-8684-e1d2735e3743, Description: Multiple Residential Accommodation, Use: Residential Accommodation, Floors: 3.0, Area (m2): 1187.012 4. ID: 0e2a2d70-f92e-4989-875d-5b6262c64495, Description: Multiple Residential Accommodation, Use: Residential Accommodation, Floors: nan, Area (m2): 123.09 5. ID: 1ca7f086-c947-4184-bae1-2fd8bde604f4, Description: Multiple Residential Accommodation, Use: Residential Accommodation, Floors: 3.0, Area (m2): 149.138

An interactive folium map has been generated with the name: exeter_buildings_with_basement_flats_map and description: Folium map showing all buildings with basement flats in Exeter, with a summary popup and tooltips for each building.

Map file: exeter_buildings_with_basement_flats_map.html

Tool Calls and Code

Tool: send_message

- Args: {'target': 'planning_agent', 'task_description': 'User wants to show buildings that have a basement flat in Exeter. Provide the general steps to solve this geospatial query.'}
- Args: {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

- Args: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_polygon' to find the specific area polygon for Exeter. Save the filtered results as 'filtered_exeter_polygon'."}
- Args: {'target': 'buildings', 'task_description': "Search for buildings with basement flats within the area defined by the artifact 'filtered_exeter_polygon'."}
- Args: {'target': 'data_analysis_agent', 'task_description': "From the artifact 'buildings_exeter_all_buildings_within_polygon', filter and return only those buildings where 'basement_presence_selfcontained' indicates the presence of a self-contained flat in the basement. Save the filtered results as 'exeter_buildings_with_basement_flats'."}
- Args: {'target': 'plotting_agent', 'task_description': "Plot the buildings with basement flats in Exeter using the artifact 'exeter_buildings_with_basement_flats'. Add a summary to the map."}

Tool: call_os_ngd

- Args: {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_polygon'}
- Args: {'filters': None, 'bbox': 'filtered_exeter_polygon', 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_all_buildings_within_polygon'}

Tool: generate_metadata_for_artifacts

- Args: {'artifact_names': ['exeter_polygon']}
- Args: {'artifact_names': ['buildings_exeter_all_buildings_within_polygon']}
- Args: {'artifact_names': ['exeter_buildings_with_basement_flats']}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_polygon(data: list):
    gdf = data[0]
    # Filter for Exeter using multiple relevant columns
    # We look for 'Exeter' in name1_text and also ensure the descriptiongroup and description are relevant
    filtered = gdf[(gdf['name1_text'].str.lower() == 'exeter') &
                  (gdf['descriptiongroup'].isin(['Settlement', 'Land Name', 'Other Name', 'Part Of Settlement', 'City', 'Town'])) &
                  (gdf['description'].str.lower().isin(['city', 'town', 'suburban area', 'named other geographic area', 'part of settlement']))]
    summary = f"Filtered Exeter polygon: {len(filtered)} entries found."
    artifact_name = 'filtered_exeter_polygon'
    artifact_description = 'GeoDataFrame containing the polygon(s) specifically for Exeter, filtered by name and relevant description columns.'
    return [summary, artifact_name, artifact_description, filtered]

import matplotlib
```

```

matplotlib.use('Agg')
def filter_exeter_buildings_with_basement_flats(data: list):
    gdf = data[0]
    # Filter for buildings where 'basementpresence_selfcontained' is 'Present'
    filtered = gdf[gdf['basementpresence_selfcontained'] == 'Present']
    summary = f"Filtered buildings with self-contained basement flats: {len(filtered)} entries."
    artifact_name = 'exeter_buildings_with_basement_flats'
    artifact_description = (
        "GeoDataFrame of buildings in Exeter where the 'basementpresence_selfcontained' column "
        "indicates the presence of a self-contained flat in the basement. Includes all original columns and geometry."
    )
    return [summary, artifact_name, artifact_description, filtered]

import matplotlib
matplotlib.use('Agg')
def plot_buildings_with_basement_flats(data: list):
    import folium
    import geopandas as gpd
    import pandas as pd

    gdf = data[0].copy()
    # Ensure CRS is EPSG:4326 for folium
    if gdf.crs is not None and gdf.crs.to_epsg() != 4326:
        gdf = gdf.to_crs(epsg=4326)

    # Convert all datetime columns to string to avoid JSON serialization issues
    for col in gdf.columns:
        if pd.api.types.is_datetime64_any_dtype(gdf[col]):
            gdf[col] = gdf[col].astype(str)

    # Create folium map centered on Exeter (approximate center from data)
    centroid = gdf.geometry.centroid.unary_union.centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

    # Add all buildings to the map
    folium.GeoJson(
        gdf,
        name='Buildings with Basement Flats',
        tooltip=folium.GeoJsonTooltip(fields=['osid', 'description', 'buildinguse', 'numberoffloors', 'geometry_area_m2'],
                                      aliases=['ID', 'Description', 'Use', 'Floors', 'Area (m2)'])
    ).add_to(m)

    # Add a summary as a folium marker
    first5 = gdf[['osid', 'description', 'buildinguse', 'numberoffloors', 'geometry_area_m2']].head(5)
    summary_html = f"<b>Buildings with Basement Flats in Exeter</b><br>Total: {len(gdf)}<br><br>First 5:<br>"
    for idx, row in first5.iterrows():
        summary_html += f"<b>ID:</b> {row['osid']}<br><b>Description:</b> {row['description']}<br><b>Use:</b> {row['buildinguse']}<br><b>Floors:</b> {row['numb"
    folium.Marker(

```

```
        location=[centroid.y, centroid.x],
        popup=folium.Popup(summary_html, max_width=400),
        icon=folium.Icon(color='blue', icon='info-sign')
    ).add_to(m)

# Save map
map_filename = 'exeter_buildings_with_basement_flats_map.html'
m.save(map_filename)

summary = f"Plotted {len(gdf)} buildings with basement flats in Exeter. First 5 entries: {first5.to_dict(orient='records')}}"
artifact_name = 'exeter_buildings_with_basement_flats_map'
artifact_description = 'Folium map showing all buildings with basement flats in Exeter, with a summary popup and tooltips for each building.'
return [summary, artifact_name, artifact_description, map_filename]
```